

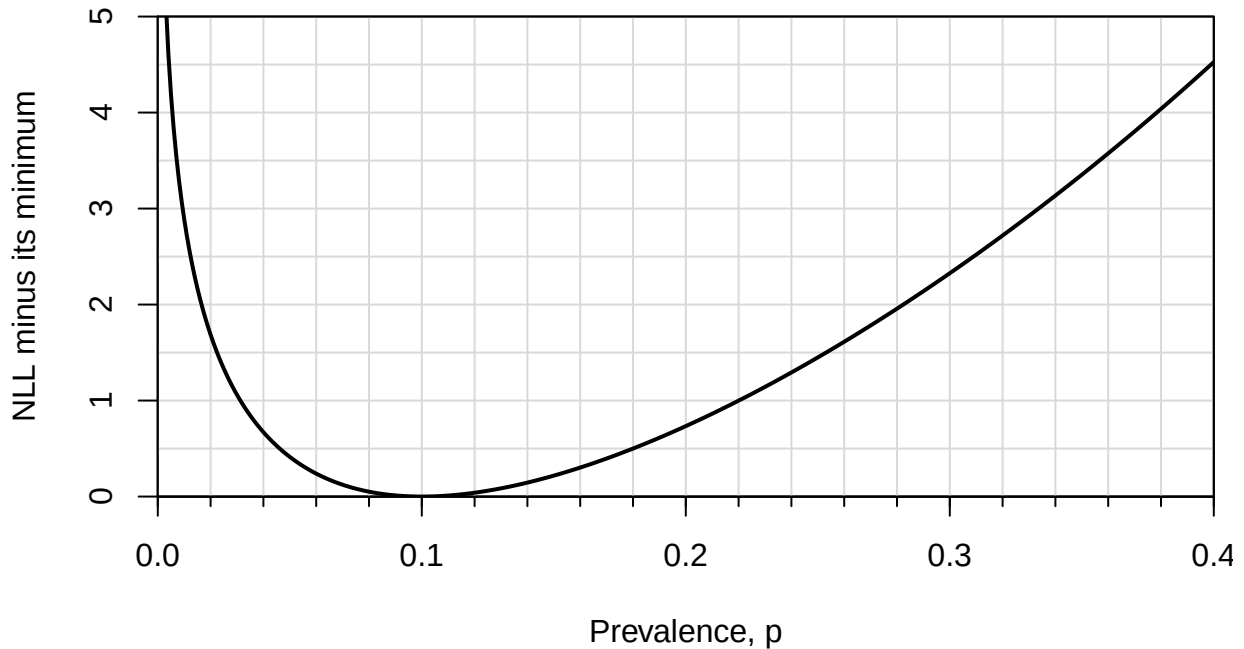
NRES 746, Week 7, Lecture 1: Reading the Interval

Name: _____

Instructions: Work alone on Part 1 (about 4 minutes), then compare with a neighbor for Part 2. Keep laptops closed.

The data

You swab 20 frogs from a pond for chytrid fungus, and 2 test positive. Model the number of positives as binomial with unknown prevalence p . The figure shows the negative log-likelihood (NLL) for these data, minus its minimum value (so the lowest point of the curve is at zero).



Part 1: Read the curve (individual)

1a. What is the maximum likelihood estimate of p ?

$\hat{p} \approx$ _____

1b. A common rule of thumb treats every value of p whose NLL is within 2 units of the minimum as plausible. Draw that cutoff on the figure, then read off the approximate range of plausible values.

From _____ to _____

1c. Is this range *symmetric* around $\hat{p}$? Why might the curve be lopsided for these data?

Part 2: More frogs (with a neighbor)

2a. A second survey swabs 80 frogs and finds 8 positive. Sketch its curve on the same figure. Will its range of plausible values be wider or narrower than yours, and does $\hat{p}$ change? Compare sketches with your neighbor.
